

14th Week Summary (12/3/26)

- **Regression**

A tool to describe how a response (dependent) variable y changes based on an explanatory (independent) variable x . Often, a straight line is used to model and predict y given x .

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i,$$

where ϵ_i are normally distributed error terms with mean 0 and constant variance σ^2 .

- **Least Squares Estimation:**

The slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$ are chosen to minimize the sum of squared residuals, $\sum (y_i - \hat{y}_i)^2$.

- **Assumptions:**

Linear relationship between x and y .

Independent, normally distributed error terms with constant variance.

Watch for outliers or influential points that may distort the regression.

- **ANOVA and R^2 :**

An ANOVA table splits total variation into that explained by the regression and that attributed to error. The coefficient of determination R^2 measures the proportion of the variation in y explained by the model.

- **Inference:**

Hypothesis tests (e.g., $H_0 : \beta_1 = 0$) determine whether there is a statistically significant linear relationship. Confidence and prediction intervals provide estimates for the mean response and for individual new observations.

- **Cautions:**

Regression lines only capture linear patterns; always check scatterplots.

Extrapolating far beyond observed x -values can be unreliable.

Residual plots are crucial for checking model assumptions.

